Caden Phillips

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MPhys Astrophysics student at the University of Liverpool, specialising in compact objects, computational modelling and time-domain astrophysics. Currently using multi-epoch spectroscopy to search for neutron stars in binaries; previous research in Fisher matrix forecasting, TOV solvers, X-ray time-domain analysis of NICER data and spectral fitting of XMM-Newton observations. Experienced in building end-to-end analysis workflows for large data sets, with strong Python and scientific computing skills. Seeking a PhD on binary stars and hidden compact objects in massive star clusters.

EDUCATION

MPhys Astrophysics, University of Liverpool *Expected 2027*
Projected First Class · Year 3: 73.5%
Academic Advisor: Dr. Christopher Copperwheat

Selected modules: Numerical Methods in Physics, Relativity and Cosmology, Advanced Observational Astrophysics, The Interstellar Medium, Physics of Planets.
Full transcript available upon request.

MPHYS PROJECT

Unlikely Friends: The Hunt for Neutron Stars with Stellar Companions *Oct 2026 – Present*
Supervisor: Dr. Rhys Seeburger

- Investigating candidate neutron star binaries using high-resolution, multi-epoch FEROS and UVES spectroscopic observations.
- Measuring stellar radial velocities and fitting binary orbits to constrain orbital parameters and assess existing Gaia orbital solutions.
- Applying spectral disentangling techniques to identify luminous secondary companions, distinguish genuine compact-object companions from imposter systems, and place upper limits on secondary-star contributions where no companion is detected.
- Potential extensions include spectroscopic determination of stellar parameters, stellar evolutionary modelling to constrain the nature of the secondary, and reconstruction of the systems' evolutionary histories.

OTHER RESEARCH EXPERIENCE

Independent Study — X-ray Weak Active Galactic Nuclei *Summer 2026*
Self-directed project based on the LEAPS (Leiden/ESA) internship; shortlisted for interview

- Built an end-to-end Python pipeline automating XMM-Newton EPIC data reduction (SAS), spectral fitting (PyXSPEC) and Bayesian hardness-ratio estimation (BEHR), applied to 90+ quasars at $z = 1.8$–2.2 drawn from a crossmatch of the 5XMM-DR15 and SDSS DR16Q catalogues.
- Classified sources as X-ray weak, normal or strong via $\Delta\alpha_{\rm ox}$, and examined how photon index, hardness ratio and fractional variability vary across the $\Delta\alpha_{\rm ox}$ distribution.
- Validated the pipeline on well-studied sources against published data.
- Compared target outputs against the 5XMM-DR15 catalogue.
- Found weak to no correlation between photon index (Γ) and $\Delta\alpha_{\rm ox}$.

Independent Study — HEASARC/NICER Analysis Pipeline *Winter 2025 – Spring 2026*
Self-directed observational analysis project

- Developed a Python pipeline integrating HEASoft tools via Bash to automatically download and organise archival NICER X-ray observations from the HEASARC database.
- Automated event filtering and calibration with scripted XSELECT workflows before light-curve extraction, across up to hundreds of ObsIDs.
- Applied the pipeline to archival observations of neutron stars.
- Produced calibrated X-ray light curves and diagnostic plots to enable time-domain comparison of source variability across observing epochs.
- Stacked light curves across epochs and flagged flares via a count-rate threshold.

Internship — Institute of Nuclear Physics (IFJ PAN), Kraków, Poland *Jul 2025 – Aug 2025*
Supervisor: Dr. David Álvarez Castillo

- Applied a Fisher information matrix framework to estimate tidal deformability uncertainties and assess error propagation in binary neutron star systems.
- Developed and implemented a Tolman–Oppenheimer–Volkoff (TOV) solver to compute mass–radius and tidal deformability relations for 62 hybrid equations of state.
- Compared model predictions with the GW170817 50% and 90% credible regions; none of the 62 hybrid EoS fell within either.
- Co-presented results at the IFJ PAN Summer Internship Symposium 2025 with a fellow intern.

Fieldwork — Teide Observatory, Tenerife, Spain *Jun 2025*
Supervisor: Dr. Daniel Harman

- Operated the 0.8 m IAC-80 telescope and a Celestron C14 Edge HD system. Identified targets and analysed the observations using Hertzsprung–Russell diagrams.

RESEARCH OUTPUTS

- **Report & presentation:** *Physical Inputs for Simulations of Gravitational Wave Emissions from Compact Stars within the Einstein Telescope Detection Range.* Institute of Nuclear Physics (IFJ PAN), Kraków, 2025; presented at the IFJ PAN Summer Internship Symposium 2025. [PDF]. [Slides]. [GitHub].
- **Report:** *A Study of X-ray Properties of Active Galactic Nuclei at* $z = 1.8$–2.2 *across the* $\Delta\alpha_{ox}$ *Spectrum.* Independent Study, 2026. [PDF]. [GitHub].
- **Code:** *Using HEASoft to Analyse Archived NICER Data of Neutron Stars.* Independent Study, 2026. [GitHub].

TECHNICAL SKILLS

- **Programming:** Python (NumPy, SciPy, Matplotlib, Astropy), C++, Bash.
- **Tools & software:** HEASoft, XMM-Newton SAS, XSPEC/PyXSPEC, BEHR, RNS, LORENE, ROOT, Jupyter, Git, LaTeX.
- **Scientific methods:** TOV and hybrid EoS modelling, Fisher matrix forecasting, spectral fitting, light-curve analysis, numerical ODE solvers.

AWARDS & MEMBERSHIPS

- Awarded a competitive summer internship placement at IFJ PAN (Poland) through the PHYS309 Internship Module. [PDF].
- Associate Member, Institute of Physics (IOP) *since 2023*.